

1.3 Inverses and generalized inverses

1.3.1 The inverse of a square matrix

An $n \times n$ matrix $\mathbf{A}$ is **nonsingular** if there is a matrix $\mathbf{A}^{-1}$ with $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$. The inverse is unique: if $\mathbf{B}$ and $\mathbf{C}$ both qualify, then $\mathbf{B} = \mathbf{B}(\mathbf{A}\mathbf{C}) = (\mathbf{B}\mathbf{A})\mathbf{C} = \mathbf{C}$.

Proposition 1.3.1

For an $n \times n$ matrix $\mathbf{A}$ the following are equivalent: (a) $\mathbf{A}$ is nonsingular; (b) $\text{rank}(\mathbf{A}) = n$; (c) $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$; (d) $\mathbf{B}\mathbf{A} = \mathbf{I}$ for some $\mathbf{B}$; (e) $\mathbf{A}\mathbf{C} = \mathbf{I}$ for some $\mathbf{C}$. When they hold, $\mathbf{B} = \mathbf{C} = \mathbf{A}^{-1}$, and

$$(\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top, \quad (\mathbf{A}\mathbf{B})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

for nonsingular $\mathbf{A}, \mathbf{B}$ of the same size. A nonsingular symmetric matrix has a symmetric inverse.

Proof. (b) $\Leftrightarrow$ (c) is rank-nullity. (a) $\Rightarrow$ (d) and (a) $\Rightarrow$ (e) are trivial. (d) $\Rightarrow$ (c): $\mathbf{A}\mathbf{x} = \mathbf{0}$ gives $\mathbf{x} = \mathbf{B}\mathbf{A}\mathbf{x} = \mathbf{0}$. (e) $\Rightarrow$ (b): $\mathbf{I} = \mathbf{A}\mathbf{C}$ has rank n , and $\text{rank}(\mathbf{A}\mathbf{C}) \leq \text{rank}(\mathbf{A})$. (b) $\Rightarrow$ (a): since $\mathcal{C}(\mathbf{A}) = \mathbb{R}^n$, we can solve $\mathbf{A}\mathbf{c}_j = \mathbf{e}_j$ for each j , giving $\mathbf{A}\mathbf{C} = \mathbf{I}$. Then $\mathbf{A}(\mathbf{C}\mathbf{A} - \mathbf{I}) = \mathbf{0}$, and because $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$ each column of $\mathbf{C}\mathbf{A} - \mathbf{I}$ is zero. So $\mathbf{C}\mathbf{A} = \mathbf{I}$ as well. If $\mathbf{B}\mathbf{A} = \mathbf{I}$ then $\mathbf{B} = \mathbf{B}\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}$, and similarly for $\mathbf{C}$. The formulas are verified by multiplication: $\mathbf{A}^\top(\mathbf{A}^{-1})^\top = (\mathbf{A}^{-1}\mathbf{A})^\top = \mathbf{I}$ and $(\mathbf{A}\mathbf{B})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{I}$. If $\mathbf{A} = \mathbf{A}^\top$, then $\mathbf{A}^{-1} = (\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top$. ■

A rectangular matrix has no two-sided inverse. If $\mathbf{X}$ is $n \times p$ with full column rank, then $\mathbf{X}^\top\mathbf{X}$ is nonsingular by Proposition 1.2.3(c), and $(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ is a *left inverse*: it multiplies $\mathbf{X}$ on the left to give $\mathbf{I}_p$. This is the matrix that computes least squares coefficients.

1.3.2 Generalized inverses

When $\mathbf{A}$ is singular or rectangular we still want a matrix that inverts $\mathbf{A}$ on the vectors where inversion makes sense, namely those in $\mathcal{C}(\mathbf{A})$. The following weak requirement turns out to be exactly what solving equations needs.

Definition 1.3.1 — Generalized inverse

A **generalized inverse** of an $m \times n$ matrix $\mathbf{A}$ is any $n \times m$ matrix $\mathbf{A}^-$ with

$$\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}.$$

Read columnwise, the definition says that $\mathbf{A}\mathbf{A}^-$ acts as the identity on every column of $\mathbf{A}$, hence on all of $\mathcal{C}(\mathbf{A})$: if $\mathbf{u} = \mathbf{A}\mathbf{b}$ then $\mathbf{A}(\mathbf{A}^-\mathbf{u}) = \mathbf{A}\mathbf{A}^-\mathbf{A}\mathbf{b} = \mathbf{u}$. So $\mathbf{A}^-\mathbf{u}$ is a solution of $\mathbf{A}\mathbf{x} = \mathbf{u}$ whenever one exists. If $\mathbf{A}$ is nonsingular, multiplying the definition by $\mathbf{A}^{-1}$ on both sides gives

$\mathbf{A}^- = \mathbf{A}^{-1}$. In general there are many generalized inverses. Chapter 6 uses them to write projections, and Chapter 8 uses them to describe estimable functions in models of less than full rank.

Theorem 1.3.2 — Existence

Every matrix has a generalized inverse. Explicitly, if $\text{rank}(\mathbf{A}) = r \geq 1$ and $\mathbf{A}_{IJ}$ is the nonsingular $r \times r$ submatrix of Proposition 1.2.4, the $n \times m$ matrix $\mathbf{G}$ that holds $\mathbf{A}_{IJ}^{-1}$ in the rows indexed by J and the columns indexed by I , and zeros elsewhere, is a generalized inverse of $\mathbf{A}$.

Proof. For $\mathbf{A} = \mathbf{0}$ any matrix of the right size works. Otherwise let $\mathbf{E}_I$ and $\mathbf{E}_J$ be the columns of $\mathbf{I}_m$ and $\mathbf{I}_n$ indexed by I and J . Then $\mathbf{G} = \mathbf{E}_J \mathbf{A}_{IJ}^{-1} \mathbf{E}_I^\top$, $\mathbf{A} \mathbf{E}_J = \mathbf{A}_{.J}$ and $\mathbf{E}_I^\top \mathbf{A} = \mathbf{A}_{I.}$. So $\mathbf{A} \mathbf{G} \mathbf{A} = \mathbf{A}_{.J} \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I.} = \mathbf{A}$ by Eq. 1.2.1. ■

Proposition 1.3.3 — Properties of generalized inverses

Let $\mathbf{A}$ be $m \times n$ of rank r and let $\mathbf{G}$ be a generalized inverse of $\mathbf{A}$.

- (a) $\mathbf{A} \mathbf{G}$ and $\mathbf{G} \mathbf{A}$ are idempotent. Moreover $\mathcal{C}(\mathbf{A} \mathbf{G}) = \mathcal{C}(\mathbf{A})$ and $\mathcal{N}(\mathbf{G} \mathbf{A}) = \mathcal{N}(\mathbf{A})$, and $\text{rank}(\mathbf{A} \mathbf{G}) = \text{rank}(\mathbf{G} \mathbf{A}) = r \leq \text{rank}(\mathbf{G})$.
- (b) $\mathbf{G}^\top$ is a generalized inverse of $\mathbf{A}^\top$.
- (c) If $\mathbf{G}_1, \mathbf{G}_2$ are generalized inverses of $\mathbf{A}$, so is $\mathbf{G}_1 \mathbf{A} \mathbf{G}_2$. In particular $\mathbf{G} \mathbf{A} \mathbf{G}$ is a generalized inverse which also satisfies $(\mathbf{G} \mathbf{A} \mathbf{G}) \mathbf{A} (\mathbf{G} \mathbf{A} \mathbf{G}) = \mathbf{G} \mathbf{A} \mathbf{G}$.
- (d) If $\mathbf{A}$ is symmetric, then $\mathbf{G}^\top$ and $\frac{1}{2}(\mathbf{G} + \mathbf{G}^\top)$ are generalized inverses of $\mathbf{A}$, so a symmetric generalized inverse always exists.
- (e) $\mathbf{G} + (\mathbf{I} - \mathbf{G} \mathbf{A}) \mathbf{U} + \mathbf{V} (\mathbf{I} - \mathbf{A} \mathbf{G})$ is a generalized inverse of $\mathbf{A}$ for every $\mathbf{U}$ and $\mathbf{V}$ of size $n \times m$.

Proof.

- (a) $(\mathbf{A} \mathbf{G})^2 = (\mathbf{A} \mathbf{G} \mathbf{A}) \mathbf{G} = \mathbf{A} \mathbf{G}$, and likewise for $\mathbf{G} \mathbf{A}$. Next, $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{A} \mathbf{G} \mathbf{A}) \subseteq \mathcal{C}(\mathbf{A} \mathbf{G}) \subseteq \mathcal{C}(\mathbf{A})$, and $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{G} \mathbf{A}) \subseteq \mathcal{N}(\mathbf{A} \mathbf{G} \mathbf{A}) = \mathcal{N}(\mathbf{A})$. The ranks follow from these and rank-nullity, and $r = \text{rank}(\mathbf{A} \mathbf{G}) \leq \text{rank}(\mathbf{G})$.
- (b) Transpose $\mathbf{A} \mathbf{G} \mathbf{A} = \mathbf{A}$. (c) $\mathbf{A} \mathbf{G}_1 \mathbf{A} \mathbf{G}_2 \mathbf{A} = \mathbf{A} \mathbf{G}_2 \mathbf{A} = \mathbf{A}$. The second statement is $\mathbf{G} \mathbf{A} \mathbf{G} \mathbf{A} \mathbf{G} = \mathbf{G} (\mathbf{A} \mathbf{G} \mathbf{A}) \mathbf{G} = \mathbf{G} \mathbf{A} \mathbf{G} = \mathbf{G} \mathbf{A} \mathbf{G}$.
- (c) By (b), $\mathbf{G}^\top$ is a generalized inverse of $\mathbf{A}^\top = \mathbf{A}$. The condition $\mathbf{A} \mathbf{G} \mathbf{A} = \mathbf{A}$ is linear in $\mathbf{G}$, so it is preserved by averaging.
- (d) Multiply out, using $\mathbf{A} (\mathbf{I} - \mathbf{G} \mathbf{A}) = \mathbf{0}$ and $(\mathbf{I} - \mathbf{A} \mathbf{G}) \mathbf{A} = \mathbf{0}$. ■

A generalized inverse with the extra property in (c), $\mathbf{G} \mathbf{A} \mathbf{G} = \mathbf{G}$, is called **reflexive**. Exercise (B1) shows that the reflexive ones are exactly those of rank r . Part (e) in fact produces *every* generalized inverse from a single one as $\mathbf{U}$ and $\mathbf{V}$ vary. Idempotence in (a) means that $\mathbf{A} \mathbf{G}$ and $\mathbf{G} \mathbf{A}$ are projections, generally not orthogonal ones. Chapter 6 shows that the particular

combination $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-}\mathbf{X}^\top$ is symmetric, does not depend on the choice of generalized inverse, and is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$.

1.3.3 The Moore–Penrose inverse

One generalized inverse is singled out by symmetry requirements.

Definition 1.3.2 — Moore–Penrose inverse

The **Moore–Penrose inverse** of $\mathbf{A}$ is the matrix $\mathbf{A}^+$ satisfying

$$\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}, \quad \mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+, \quad (\mathbf{A}\mathbf{A}^+)^\top = \mathbf{A}\mathbf{A}^+, \quad (\mathbf{A}^+\mathbf{A})^\top = \mathbf{A}^+\mathbf{A}.$$

There is at most one such matrix. If $\mathbf{G}$ and $\mathbf{H}$ both satisfy the four conditions, then

$$\mathbf{G} = \mathbf{G}\mathbf{A}\mathbf{G} = \mathbf{G}(\mathbf{A}\mathbf{G})^\top = \mathbf{G}\mathbf{G}^\top(\mathbf{A}\mathbf{H}\mathbf{A})^\top = \mathbf{G}(\mathbf{A}\mathbf{G})^\top(\mathbf{A}\mathbf{H})^\top = \mathbf{G}\mathbf{A}\mathbf{G}\mathbf{A}\mathbf{H} = \mathbf{G}\mathbf{A}\mathbf{H},$$

where the steps use the second and third conditions for $\mathbf{G}$, the first for $\mathbf{H}$, and the third for both. The same argument with the fourth condition in place of the third gives

$$\mathbf{H} = (\mathbf{H}\mathbf{A})^\top \mathbf{H} = \mathbf{A}^\top \mathbf{H}^\top \mathbf{H} = (\mathbf{A}\mathbf{G}\mathbf{A})^\top \mathbf{H}^\top \mathbf{H} = (\mathbf{G}\mathbf{A})^\top (\mathbf{H}\mathbf{A})^\top \mathbf{H} = \mathbf{G}\mathbf{A}\mathbf{H}\mathbf{A}\mathbf{H} = \mathbf{G}\mathbf{A}\mathbf{H},$$

so $\mathbf{G} = \mathbf{H}$. Existence follows from the singular value decomposition, which gives $\mathbf{A}^+$ explicitly (Theorem 1.8.1).

1.3.4 Exercises

1.3.4.1 A. Check your understanding

Exercise A1.

Let $\mathbf{A} = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{pmatrix}$. (a) Check that $\text{rank}(\mathbf{A}) = 1$ and use the recipe of [Theorem 1.3.2](#) with $I = \{1\}$ and $J = \{1\}$ to write down a generalized inverse $\mathbf{G}$; verify $\mathbf{A}\mathbf{G}\mathbf{A} = \mathbf{A}$. (b) Show that $\mathbf{A}\mathbf{G}$ is not symmetric, so $\mathbf{G} \neq \mathbf{A}^+$, although $\mathbf{G}\mathbf{A}\mathbf{G} = \mathbf{G}$. (c) Verify that $\mathbf{A}^+ = \frac{1}{70} \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ 3 & 6 \end{pmatrix}$ satisfies the four conditions of [Definition 1.3.2](#).

Solution.

- (a) The second row is twice the first, so the rank is one, and the nonsingular 1×1 submatrix at $I = J = \{1\}$ is (1) . The recipe puts 1 in the entry of the 3×2 matrix $\mathbf{G}$ indexed by row 1 and column 1, and zeros elsewhere. Then $\mathbf{A}\mathbf{G} = \begin{pmatrix} 1 & 0 \\ 2 & 0 \end{pmatrix}$ and $\mathbf{A}\mathbf{G}\mathbf{A} = \mathbf{A}$.
- (b) The matrix $\mathbf{A}\mathbf{G}$ is visibly not symmetric. And $\mathbf{G}(\mathbf{A}\mathbf{G}) = \mathbf{G}$, because $\mathbf{G}$ has only its $(1, 1)$ entry nonzero, so $\mathbf{G}(\mathbf{A}\mathbf{G})$ has the first row $(1, 0)$ of $\mathbf{A}\mathbf{G}$ as its own first row and zeros elsewhere.
- (c) Write $\mathbf{A} = \mathbf{u}\mathbf{v}^\top$ with $\mathbf{u} = (1, 2)^\top$ and $\mathbf{v} = (1, 2, 3)^\top$, so $\mathbf{u}^\top \mathbf{u} = 5$, $\mathbf{v}^\top \mathbf{v} = 14$ and $\mathbf{A}^+ = \mathbf{v}\mathbf{u}^\top/70$. Then $\mathbf{A}\mathbf{A}^+ = \mathbf{u}\mathbf{u}^\top/5$ and $\mathbf{A}^+\mathbf{A} = \mathbf{v}\mathbf{v}^\top/14$, both symmetric;

$$\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{u}(\mathbf{u}^\top \mathbf{u})(\mathbf{v}^\top \mathbf{v})\mathbf{v}^\top / 70 = \mathbf{u}\mathbf{v}^\top = \mathbf{A}; \text{ and } \mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{v}(\mathbf{v}^\top \mathbf{v})(\mathbf{u}^\top \mathbf{u})\mathbf{u}^\top / 70^2 = \mathbf{A}^+.$$

Exercise A2.

Let $\mathbf{A} = \begin{pmatrix} \mathbf{I}_r & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ be $m \times n$. Show that the generalized inverses of $\mathbf{A}$ are exactly the $n \times m$ matrices $\mathbf{G} = \begin{pmatrix} \mathbf{I}_r & \mathbf{U} \\ \mathbf{V} & \mathbf{W} \end{pmatrix}$ with $\mathbf{U}, \mathbf{V}, \mathbf{W}$ arbitrary; that $\mathbf{G}$ is reflexive iff $\mathbf{W} = \mathbf{V}\mathbf{U}$; and that $\mathbf{A}^+ = \mathbf{A}^\top$. How many generalized inverses does this $\mathbf{A}$ have when $r < \min(m, n)$, and how many when $r = m = n$?

Solution. Partition $\mathbf{G}$ conformably. Block multiplication gives $\mathbf{A}\mathbf{G}\mathbf{A} = \begin{pmatrix} \mathbf{G}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$, which equals $\mathbf{A}$ iff $\mathbf{G}_{11} = \mathbf{I}_r$; the other blocks are unconstrained. With $\mathbf{G}_{11} = \mathbf{I}_r$, $\mathbf{G}\mathbf{A} = \begin{pmatrix} \mathbf{I}_r & \mathbf{0} \\ \mathbf{V} & \mathbf{0} \end{pmatrix}$ and $\mathbf{G}\mathbf{A}\mathbf{G} = \begin{pmatrix} \mathbf{I}_r & \mathbf{U} \\ \mathbf{V} & \mathbf{V}\mathbf{U} \end{pmatrix}$, so $\mathbf{G}\mathbf{A}\mathbf{G} = \mathbf{G}$ iff $\mathbf{W} = \mathbf{V}\mathbf{U}$. For the Moore–Penrose inverse, $\mathbf{A}\mathbf{G}$ and $\mathbf{G}\mathbf{A}$ are symmetric only if $\mathbf{U} = \mathbf{0}$ and $\mathbf{V} = \mathbf{0}$, and then reflexivity forces $\mathbf{W} = \mathbf{0}$. The remaining matrix is $\mathbf{A}^\top$. When $r < \min(m, n)$ at least one of $\mathbf{U}, \mathbf{V}, \mathbf{W}$ is a nonempty block, so there are infinitely many generalized inverses; when $r = m = n$ the three blocks are empty and $\mathbf{G} = \mathbf{I}_n = \mathbf{A}^{-1}$ is the only one, as it must be.

Exercise A3.

Show that a symmetric idempotent matrix is its own Moore–Penrose inverse, and deduce $(\mathbf{I}_n - n^{-1}\mathbf{J}_n)^+ = \mathbf{I}_n - n^{-1}\mathbf{J}_n$ for the centring matrix of Example (Sums of squares as quadratic forms). Show also that $\mathbf{J}_n^+ = n^{-2}\mathbf{J}_n$.

1.3.4.2 B. Practice**Exercise B1.**

Let $\mathbf{X}$ be $n \times p$ with full column rank and write $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$. Show that the left inverses of $\mathbf{X}$, that is, the $p \times n$ matrices $\mathbf{L}$ with $\mathbf{L}\mathbf{X} = \mathbf{I}_p$, are exactly

$$\mathbf{L} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top + \mathbf{K}(\mathbf{I}_n - \mathbf{M}), \quad \mathbf{K} \text{ arbitrary,}$$

that every left inverse is a generalized inverse of $\mathbf{X}$, and that $(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$ is the only left inverse whose rows lie in $\mathcal{C}(\mathbf{X})$.

Solution. Since $\mathbf{M}\mathbf{X} = \mathbf{X}$, every matrix of the stated form satisfies $\mathbf{L}\mathbf{X} = \mathbf{I}_p + \mathbf{K}(\mathbf{X} - \mathbf{X}) = \mathbf{I}_p$. Conversely, if $\mathbf{L}\mathbf{X} = \mathbf{I}_p$, then $\mathbf{L} = \mathbf{L}\mathbf{M} + \mathbf{L}(\mathbf{I}_n - \mathbf{M}) = (\mathbf{L}\mathbf{X})(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top + \mathbf{L}(\mathbf{I}_n - \mathbf{M})$, which is of the stated form with $\mathbf{K} = \mathbf{L}$. A left inverse satisfies $\mathbf{X}\mathbf{L}\mathbf{X} = \mathbf{X}$, so it is a generalized inverse. The rows of $(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$ are the vectors $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{e}_j$, which lie in $\mathcal{C}(\mathbf{X})$; the rows of $\mathbf{K}(\mathbf{I}_n - \mathbf{M})$ are of the form $(\mathbf{I}_n - \mathbf{M})\mathbf{c}$, and $\mathbf{X}^\top(\mathbf{I}_n - \mathbf{M}) = \mathbf{0}$ shows that such a vector lies in $\mathcal{C}(\mathbf{X})$ only if it is zero, since $\mathbf{X}\mathbf{b} = (\mathbf{I}_n - \mathbf{M})\mathbf{c}$ gives $\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{0}$ and hence $\mathbf{X}\mathbf{b} = \mathbf{0}$ by Exercise (A2).

Exercise B2.

Let $\mathbf{X}$ be $n \times p$ of any rank and let $\mathbf{G}$ be any generalized inverse of $\mathbf{X}^\top \mathbf{X}$. Show that (a) $\mathbf{X}\mathbf{G}\mathbf{X}^\top \mathbf{X} = \mathbf{X}$, so $\mathbf{G}\mathbf{X}^\top$ is a generalized inverse of $\mathbf{X}$; (b) $\mathbf{X}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top = \mathbf{X}^\top$; and (c)

$\mathbf{M} = \mathbf{XGX}^\top$ is symmetric and idempotent and is the same matrix for every choice of $\mathbf{G}$. *Hints:* for (a) use Exercise (A2); for (b) apply (a) to $\mathbf{G}^\top$, which is also a generalized inverse of $\mathbf{X}^\top \mathbf{X}$ by Proposition 1.3.3(b). The matrix $\mathbf{M}$ is the orthogonal projection onto $\mathcal{C}(\mathbf{X})$ of Chapter 6, and this exercise is what makes it well defined when $\mathbf{X}$ has less than full rank.

Solution.

- (a) Because $\mathbf{G}$ is a generalized inverse of $\mathbf{X}^\top \mathbf{X}$, we have $\mathbf{X}^\top \mathbf{X}(\mathbf{GX}^\top \mathbf{X}) = \mathbf{X}^\top \mathbf{X} \mathbf{I}_p$. Cancelling one factor $\mathbf{X}^\top$ on the left, which Exercise (A2) permits, gives $\mathbf{XGX}^\top \mathbf{X} = \mathbf{X}$. Hence $\mathbf{X}(\mathbf{GX}^\top) \mathbf{X} = \mathbf{X}$.
- (b) Since $\mathbf{X}^\top \mathbf{X}$ is symmetric, $\mathbf{G}^\top$ is a generalized inverse of it, so (a) gives $\mathbf{XG}^\top \mathbf{X}^\top \mathbf{X} = \mathbf{X}$; transposing yields $\mathbf{X}^\top \mathbf{XGX}^\top = \mathbf{X}^\top$.
- (c) Write $\mathbf{M} = \mathbf{XGX}^\top$, so $\mathbf{M}^\top = \mathbf{XG}^\top \mathbf{X}^\top$. Using (b), $\mathbf{M}^\top \mathbf{M} = \mathbf{XG}^\top (\mathbf{X}^\top \mathbf{XGX}^\top) = \mathbf{XG}^\top \mathbf{X}^\top = \mathbf{M}^\top$. The left-hand side is symmetric, so $\mathbf{M}^\top$ is symmetric, that is, $\mathbf{M}^\top = \mathbf{M}$; and then $\mathbf{M}^2 = \mathbf{M}^\top \mathbf{M} = \mathbf{M}$. If $\mathbf{G}_1, \mathbf{G}_2$ are two generalized inverses of $\mathbf{X}^\top \mathbf{X}$, then by (a) for $\mathbf{G}_1$ and (b) for $\mathbf{G}_2$,

$$\mathbf{XG}_2 \mathbf{X}^\top = (\mathbf{XG}_1 \mathbf{X}^\top \mathbf{X}) \mathbf{G}_2 \mathbf{X}^\top = \mathbf{XG}_1 (\mathbf{X}^\top \mathbf{XG}_2 \mathbf{X}^\top) = \mathbf{XG}_1 \mathbf{X}^\top.$$

Exercise B3.

The text says that Proposition 1.3.3(e) produces *every* generalized inverse from a single one. Prove it: if $\mathbf{G}_0$ and $\mathbf{G}$ are generalized inverses of $\mathbf{A}$, find $\mathbf{U}$ and $\mathbf{V}$ with $\mathbf{G} = \mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0 \mathbf{A}) \mathbf{U} + \mathbf{V} (\mathbf{I} - \mathbf{A} \mathbf{G}_0)$.

Solution. Put $\mathbf{Z} = \mathbf{G} - \mathbf{G}_0$, so that $\mathbf{AZA} = \mathbf{A} - \mathbf{A} = \mathbf{0}$. Take $\mathbf{U} = \mathbf{Z}$ and $\mathbf{V} = \mathbf{G}_0 \mathbf{AZ}$. Then

$$(\mathbf{I} - \mathbf{G}_0 \mathbf{A}) \mathbf{Z} + \mathbf{G}_0 \mathbf{AZ} (\mathbf{I} - \mathbf{A} \mathbf{G}_0) = \mathbf{Z} - \mathbf{G}_0 \mathbf{AZ} + \mathbf{G}_0 \mathbf{AZ} - \mathbf{G}_0 (\mathbf{AZA}) \mathbf{G}_0 = \mathbf{Z},$$

so $\mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0 \mathbf{A}) \mathbf{U} + \mathbf{V} (\mathbf{I} - \mathbf{A} \mathbf{G}_0) = \mathbf{G}_0 + \mathbf{Z} = \mathbf{G}$.

1.3.4.3 C. Going deeper

Exercise C1.

Call $\mathbf{G}$ a **least-squares generalized inverse** of $\mathbf{A}$ if it satisfies the first and third conditions of Definition 1.3.2, namely $\mathbf{AGA} = \mathbf{A}$ and $(\mathbf{AG})^\top = \mathbf{AG}$. Show that (a) $\mathbf{G}_0 = (\mathbf{A}^\top \mathbf{A})^- \mathbf{A}^\top$ is one, for every choice of generalized inverse of $\mathbf{A}^\top \mathbf{A}$; (b) if $\mathbf{G}$ is a least-squares generalized inverse, then $\mathbf{x} = \mathbf{Gb}$ minimizes $\|\mathbf{Ax} - \mathbf{b}\|$, for every $\mathbf{b}$; and (c) conversely, if $\mathbf{Gb}$ minimizes $\|\mathbf{Ax} - \mathbf{b}\|$ for every $\mathbf{b}$, then $\mathbf{G}$ is a least-squares generalized inverse. So these are exactly the matrices that solve least squares problems, whatever the rank of $\mathbf{A}$.

Solution.

- (a) is Exercise (B2) with $\mathbf{X} = \mathbf{A}$: part (a) there gives $\mathbf{AG}_0 \mathbf{A} = \mathbf{A}$ and part (c) gives the symmetry of $\mathbf{AG}_0$.
- (b) First, $\mathbf{A}^\top (\mathbf{AG} - \mathbf{I}) = (\mathbf{AG})^\top \mathbf{A}^\top - \mathbf{A}^\top = (\mathbf{AGA})^\top - \mathbf{A}^\top = \mathbf{0}$, using the symmetry

of $\mathbf{AG}$ and then $\mathbf{AGA} = \mathbf{A}$. Hence for every $\mathbf{x}$ and $\mathbf{b}$ the cross term in

$$\|\mathbf{Ax} - \mathbf{b}\|^2 = \|\mathbf{A}(\mathbf{x} - \mathbf{Gb})\|^2 + 2(\mathbf{x} - \mathbf{Gb})^\top \mathbf{A}^\top (\mathbf{AG} - \mathbf{I})\mathbf{b} + \|\mathbf{AGb} - \mathbf{b}\|^2$$

vanishes, and $\|\mathbf{Ax} - \mathbf{b}\|^2 = \|\mathbf{A}(\mathbf{x} - \mathbf{Gb})\|^2 + \|\mathbf{AGb} - \mathbf{b}\|^2 \geq \|\mathbf{AGb} - \mathbf{b}\|^2$, with equality at $\mathbf{x} = \mathbf{Gb}$.

- (c) Let $\mathbf{G}_0$ be the matrix of (a). By (b) the minimum of $\|\mathbf{Ax} - \mathbf{b}\|$ equals $\|\mathbf{AG}_0\mathbf{b} - \mathbf{b}\|$, so the hypothesis says $\|\mathbf{AGb} - \mathbf{b}\| = \|\mathbf{AG}_0\mathbf{b} - \mathbf{b}\|$ for every $\mathbf{b}$. Applying the displayed decomposition with $\mathbf{G}_0$ in place of $\mathbf{G}$ and $\mathbf{x} = \mathbf{Gb}$,

$$\|\mathbf{AGb} - \mathbf{b}\|^2 = \|\mathbf{A}(\mathbf{G} - \mathbf{G}_0)\mathbf{b}\|^2 + \|\mathbf{AG}_0\mathbf{b} - \mathbf{b}\|^2,$$

so $\mathbf{A}(\mathbf{G} - \mathbf{G}_0)\mathbf{b} = \mathbf{0}$ for every $\mathbf{b}$, that is, $\mathbf{AG} = \mathbf{AG}_0$. Then $\mathbf{AGA} = \mathbf{AG}_0\mathbf{A} = \mathbf{A}$ and $(\mathbf{AG})^\top = (\mathbf{AG}_0)^\top = \mathbf{AG}_0 = \mathbf{AG}$.

Exercise C2.

Let $\mathbf{A}$ be $m \times k$ with full column rank and $\mathbf{B}$ be $k \times n$ with full row rank. Check the four conditions of Definition 1.3.2 to show that $\mathbf{A}^+ = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$ and $\mathbf{B}^+ = \mathbf{B}^\top (\mathbf{B} \mathbf{B}^\top)^{-1}$, and then that the *reverse-order law* $(\mathbf{AB})^+ = \mathbf{B}^+ \mathbf{A}^+$ holds. (Without the rank conditions it can fail: Exercise (C1) asks for an example.)

Solution. Both $\mathbf{A}^\top \mathbf{A}$ and $\mathbf{B} \mathbf{B}^\top$ are nonsingular by Proposition 1.2.3(c). Writing $\mathbf{G} = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$, we have $\mathbf{GA} = \mathbf{I}_k$, so $\mathbf{AGA} = \mathbf{A}$ and $\mathbf{GAG} = \mathbf{G}$, while $\mathbf{AG} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$ is symmetric and $\mathbf{GA} = \mathbf{I}_k$ is symmetric. Hence $\mathbf{G} = \mathbf{A}^+$. Transposing the four conditions for $\mathbf{A}^+$ gives the four conditions for $(\mathbf{A}^+)^\top$ as a Moore–Penrose inverse of $\mathbf{A}^\top$, so $(\mathbf{A}^\top)^+ = (\mathbf{A}^+)^\top$; applying the first formula to $\mathbf{B}^\top$, which has full column rank, gives $\mathbf{B}^+ = ((\mathbf{B}^\top)^+)^\top = \mathbf{B}^\top (\mathbf{B} \mathbf{B}^\top)^{-1}$.

Now put $\mathbf{P} = \mathbf{B}^+ \mathbf{A}^+ = \mathbf{B}^\top (\mathbf{B} \mathbf{B}^\top)^{-1} (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$. Then $\mathbf{ABP} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top$ and $\mathbf{PAB} = \mathbf{B}^\top (\mathbf{B} \mathbf{B}^\top)^{-1} \mathbf{B}$, both symmetric, because $\mathbf{B} \mathbf{B}^\top (\mathbf{B} \mathbf{B}^\top)^{-1} = \mathbf{I}$ and $(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{A} = \mathbf{I}$ collapse the middle factors. The same collapsing gives $\mathbf{ABPAB} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{AB} = \mathbf{AB}$ and $\mathbf{PABP} = \mathbf{B}^\top (\mathbf{B} \mathbf{B}^\top)^{-1} \mathbf{B} \mathbf{B}^\top (\mathbf{B} \mathbf{B}^\top)^{-1} (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top = \mathbf{P}$. All four conditions hold, so $\mathbf{P} = (\mathbf{AB})^+$.